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Kansei-driven and TRIZ-informed product service systems design with abductive logic: case study of children's digital Reading book

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ABSTRACT

To achieve successful product design in an industrial context, a holistic customer requirement and emotion intelligence-driven product design method, incorporating abductive logic, has been proposed. This comprehensive PSS design approach with abductive logic, titled '4C' model. It integrates Kansei Engineering (KE), the Theory of Inventive Problem Solving (TRIZ), and Fuzzy techniques to effectively handle design knowledge in the innovation of product conceptualisation and design. It provides a traceable path linking customer concerns, business context, and technological digitisation through abductive logic. In the post-COVID-19 era, the digital transformation trend has become mainstream in the educational sector. This shift from offline to online learning offers significant advantages in terms of speed and convenience, particularly for young students. However, online courses, study materials and digital reading product must also address the emotional needs of their young users. Consequently, effective study product service systems (PSS) design requires a distinct set of design considerations, and the exploration of effective product design or product service system (PSS) design becomes essential. Therefore, an innovative case study of children's educational product design is chosen and presented to illustrate and verify the validation of 4C model.

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1. Introduction

Product design stands as a pivotal concern within both practical and academic realms. Characterised by its creative, problem-solving, and user-centric nature, product design necessitates a multifaceted approach (Palmié et al. 2024). The design synthesis process, in particular, is an intricate and sophisticated decision-making endeavour that involves identifying problems and devising solutions (Liang, Wang, and Liu 2025). This process

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incorporates three primary forms of logical reasoning: deductive, inductive, and abductive (Kolko 2010; Lee, Chen, and Lee 2020). Deductive and inductive reasoning involve drawing conclusions from established knowledge (Lee, Chen, and Lee 2020; Mirza et al. 2014; O'Reilly 2016). In contrast, abductive reasoning explains what is known or most likely, thereby facilitating the generation of new knowledge and insights throughout the design process.

Initially, design methods were regarded as tools for understanding problems by exploring customer needs and corporate objectives through novel approaches. More recently, design methods have evolved into abductive-logic design approaches (Kolko 2010; Mirza et al. 2014; O'Reilly 2016). Designers can employ these methods to trace, map, and refine requirements. Additionally, they can be utilised to identify suitable product service system (PSS) ideas (Bertoni, Carulli, and Rovida 2009; Mont 2002), technological capabilities, and processes, particularly when confronted with the challenges of advanced technological system design (Dong, Lovallo, and Mounarath 2015; Menzies 1996; Rijdsdijk and Hultink 2009).

Analyzing problems using appropriate abductive-logic design synthesis to comprehend the nature of issues and achieve breakthrough solutions that fit the context and enhance core competitiveness remains a crucial challenge in PSS design literature (Dong, Lovallo, and Mounarath 2015; Lee et al. 2018; Maussang, Zwolinski, and Brissaud 2009). Moreover, customer requirement (CR)-centric design, which acknowledges human limitations in representing implicit customer requirements, may lead to ambiguous and uncertain domain knowledge understanding especially in the Large Language Model-driven Artificial Intelligence (LLM-driven AI) era (Lee, Chen, and Lee 2020; Yang and Xing 2013; Lee and Zhao 2024; Trappey et al. 2024). Therefore, incorporating abductive logic into the CR-centric design lifecycle can help mitigate these disadvantages during the early stages of product concept design and technology innovation conceptualisation, especially for specific groups such as children or the elderly (Dong, Lovallo, and Mounarath 2015; Lee, Chen, and Trappey 2019).

Due to increasing international rivalry, technological advancements, consumer sophistication and the highly competitive globalisation of business environment, more and more enterprises switch from only selling physical products to providing service solutions. This enables enterprises to rethink the new value proposition for customers and, moreover, to build constant distinctions and entry barriers for their competitors. In addition, by providing intangible service and not just selling the physical products, it is possible to satisfy the increasing customer demand for company support and interaction during the entire product life cycle. This awareness of reliability and innovation leads to the concept of the Product Service Systems (PSS) (Lee, Chen, and Trappey 2017, 2019; Morelli 2006; Zhou et al. 2024). Typical PSS engineering often focuses on the extension of the traditional functionality of a product by incorporating additional services. However, the development of (Bertoni, Carulli, and Rovida 2009; Lee et al. 2018; Lee, Wang, and Li 2023; Zhou et al. 2024) effective product-service combinations represents more than a service into a product, service aspects are easily managed in isolation.

Given the challenges of effectively designing a product service system (PSS) as outlined above, addressing the ever-changing emotional needs of consumers in children's digital reading book products emerges as a critical design issue (Lee, Wang, and Li 2023; Wu, Lee, and Chen 2017). These products, by their nature, integrate software systems to form a digital PSS, thereby increasing complexity. Typically, product design requirements embedded within a digital service system are diverse, and the elements provided by users cannot be

easily and accurately mapped to the specific design parameter information required by technical personnel (López-Escribano, Valverde-Montesino, and García-Ortega 2021). Consequently, effectively linking design requirements with users' emotional needs becomes paramount for the innovative design of children's digital reading book products.

To date, several researchers have investigated digital reading products for children, exploring virtual interactive experiences with e-readers and proposing that future e-readers for children should prioritise initiative and personalisation while enhancing entertainment experiences (Ghalebandi and Noorhidawati 2019). Concurrently, the application of augmented reality technology in children's 3D interactive books has been explored (Amara et al. 2023). Additionally, the importance of aligning technological advances with children's education, improving information technology literacy, limiting screen time, selecting appropriate media, and monitoring the digital environment for children has been emphasised (Park, Chang, and Park 2019; Xu et al. 2022).

While previous studies have addressed various aspects of children's digital reading book products, there remains a gap in research that adopts a product service system (PSS) perspective and an abductive logic design approach (Lee, Chen, and Lee 2020; Lee, Chen, and Trappey 2019; Lee, Wang, and Li 2023) to consider users' emotional needs and effectively integrate requirement reasoning into the design phase. To bridge these gaps, this study aims to explore an abductive PSS design approach that incorporates Kansei Engineering (KE), the Theory of Inventive Problem Solving (TRIZ), and Fuzzy techniques. An empirical case study of the service functions of children's digital reading book products is presented to verify the proposed design method. Based on the previous review, there are three research questions that we attempt to answer in this study: (1) How can Kansei Engineering (KE) be effectively integrated with the Theory of Inventive Problem Solving (TRIZ) and Fuzzy techniques to enhance the design of Product Service Systems (PSS)? (2) How can the emotional needs of users, particularly children, be incorporated into the design of digital reading products to enhance user satisfaction? (3) How can abductive logic be utilised to link customer concerns, business context, and technological digitisation in the design process?

The remainder of this paper is organised as follows: Section 2 provides an overview of the research background and reviews the relevant literature. Section 3 outlines the research framework, which integrates Kansei Engineering (KE), the Theory of Inventive Problem Solving (TRIZ), and fuzzy mathematics. Section 4 presents an empirical illustration, using a children's digital reading book product as a case study. This section demonstrates how to analyze users' emotional needs through KE, rank the importance of perceptual vocabulary using fuzzy mathematics, and combine these insights with TRIZ theory to derive product concepts and innovative solutions. The final section summarises the discussion and conclusions, highlighting the main findings, academic contributions, empirical implications, and limitations of this study.

2. Literature review

2.1. Digital reading book products for children

Digital books refer to e-books, story apps, picture book apps, and iBooks which provide fictional narratives with text, illustrations, sound, and interactive features presented on a touchscreen (Kucirkova 2019). Furenes, Kucirkova, and Bus (2021) stated that various experimental studies have focused on comparing children's reading of digital and print books,

finding that digital enhancements aligned with story content can support children's reading outcomes, and that children are more focused on reading digital than paper books. López-Escribano, Valverde-Montesino, and García-Ortega (2021) indicated that children who read digitally learn more vocabulary than those who do not, and that interactive formats in digital reading products have better outcomes than static digital products. The significance of requirement-driven approaches in education was underscored in UNESCO's action plan for the 2030 Agenda for Sustainable Development (Tosun and Mihci 2020). This plan examined parents' digital education behaviours within a human-centric framework, particularly in the context of the post-COVID-19 digital transformation era. Despite extensive research on digital reading for children, there remains a notable gap in studies that explore its design from a product service system (PSS) perspective using abductive logic.

2.2. Kansei engineering and product services service design

A Product Service System (PSS) is a novel selling concept that integrates tangible products and intangible services to meet customer needs (Oliva and Kallenberg 2003). Comprising products, services, and software (Long et al. 2020; Shimomura and Hara 2010), PSS components increase design and development complexity due to the need for integrating diverse information across various professional domains (Ren and Zheng 2024).

Kansei Engineering (KE) is a design approach that emphasises integrating emotional and sensory aspects into product development (Deng 2022; Yan and Li 2021; Zabotto et al. 2019). Guo, Liu, and Chai (2014) and Yeh and Chen (2018) highlighted the importance of customer feedback in new product service development and innovation, integrating KE with hospitality services. Shi et al. (2019) and Zhou et al. (2021) applied KE to geriatric healthcare product design, enhancing elderly care and promoting industry development. Park, Park, and Yoo (2018) and Hartono (2020) used KE to design sustainable application services, developing a KE-based approach to meet customers' emotional needs while considering social, environmental, and economic performance. Akgül et al. (2022) applied fuzzy linguistic summarisation and fuzzy association rule mining to Kansei Engineering in a case study on cradle design, demonstrating the extraction of affective needs into product design elements.

While some scholars have explored KE-based approaches, many typical PSS designs do not employ KE-based methods due to the limitations of Kansei Engineering. Long et al. (2020) explored the application of Product-Service Systems (PSSs) to Reusable Packaging Systems (RPSs) as a strategic design tool to address the packaging crisis in the food and household products industry. Their study involved case studies and expert interviews to develop a classification of 15 archetypal models and a design tool evaluated by packaging professionals. Liu, Jia, and Kong (2020) proposed a systematic framework for identifying engineering characteristics for sustainable PSSs using specific processes, methods, and big data analysis, with a case study to validate the methodology. In the context of Industry 4.0, Jiao et al. (2021) introduced the concept of Design Engineering 4.0 (DE4.0), which integrates cyber and physical technologies to facilitate customer-centric design and leverage the benefits of Industry 4.0. Mastrogiacomo et al. (2021) presented a methodology to infer quality determinants of PSSs from user-generated content, exemplified in a car-sharing PSS. Nemoto, Dhiman, and Röcker (2021) proposed a conceptual framework for designing PSSs for continuous use, aiming to enhance customer acceptance and usage

by integrating various theories and literature streams. Zhang (2022) integrated artificial intelligence with industrial design to transform traditional design modes and enhance the design process. Zhang et al. (2021) proposed a sustainable Product Service Systems (PSS) development framework based on design-centric complexity (DCC) theory to enhance PSS sustainability and support social benefits. Wang et al. (2021) proposed a graph-based context-aware requirement elicitation approach for Smart PSS, aiming to extract requirements from user-generated and product-sensed data. Additionally, Wang et al. (2021) presented a context-aware concept evaluation approach based on user experiences for Smart PSS design iteration. Tunn et al. (2021) discussed consumer adoption of access-based Product-Service Systems, highlighting the influence of usage duration and product type on consumer preferences. Carrera-Rivera, Larrinaga, and Lasa (2022) conducted a literature review on context-awareness for the design of Smart-product Service Systems. Nie, Zhang, and Wang (2023) proposed screening methods for recessive couplings and interventions for Recessive Ill-conditioned Features in the early-stage design process. Fahrudin and Wahyudi (2023) modelled inventory systems using the User Experience Design Model Method, focusing on creating a web-based inventory information system that meets user needs and provides a unique user experience.

Drawing from the research background and literature review presented above, it is evident that Kansei Engineering (KE) has progressively assumed a significant role in the design of product service systems. Nevertheless, given the diverse industries, application contexts, and varying emotional requirements of users, the manner in which KE can be effectively combined with other methods to enhance design suitability remains an open research question that warrants further discussion.

2.3. TRIZ (Theory of Inventive Problem Solving) and product services services design

TRIZ theory, proposed by Soviet inventor Altshuller in 1946, is an inventive problem-solving approach based on the study of high-level patents. It includes 40 innovation principles and 39 technical engineering parameters (Altshuller 1999) that resolve technical contradictions without compromise, guiding users to desired solutions (Altshuller 1997; Lee, Chen, and Lee 2020; Shie et al. 2021). In the literature, numerous of TRIZ was traditionally adopted in engineering fields or product design. It has been proved as the useful method in the academic and practical fields. However, TRIZ is nowadays adopted in the pure service systems (Chai, Zhang, and Tan 2005; Lee, Wang, and Trappey 2015; Shie et al. 2025), product service systems (Bertoni, Carulli, and Rovida 2009) for idea generation. Chou (2021) introduced a TRIZ-based Product-Service Design Approach for developing innovative products. Jiang et al. (2021) provided an innovation design approach for PSS based on TRIZ and Function Incentive, constructing a function model of the PSS using service blueprints and function system diagrams.

The TRIZ knowledge base can assist design teams in addressing the ill-structured nature of PSS design problems, where problem causes and solutions are open-ended. Pezzotta et al. (2011) applied a TRIZ-based approach using evolution laws in a servitization case involving copy printer machines. Bertoni, Carulli, and Rovida (2009) proposed a five-phase TRIZ module mapped to generic PSS conceptual design for anti-slip devices. However, most guidelines and principles focus on product-oriented PSS rather than use-oriented or

result-oriented PSS, which derive greater value from service or business models. From a PSS life-cycle perspective, value can stem from design for delivery, maintenance, recycling, and remanufacturing, not just usage (Pezzotta et al. 2011).

Moreover, recent applications include Tan et al. (2021) using TRIZ principles to develop a solution model for improving children's drinking habits. Spreafico (2021) applied TRIZ to enhance device sustainability by reducing environmental impacts. Chung et al. (2022) integrated TRIZ into fast-food industry concept development, demonstrating its innovation potential in service performance improvement. Lin and Chen (2021) explored TRIZ in new product development and supply chain management, achieving sustainable business alternatives even in challenging industry conditions. Notably, the literature has not yet explored using TRIZ theory as a research method for user emotion-driven approaches, such as combining Kansei Engineering with TRIZ and abductive logic to address complex PSS design requirements.

2.4. Fuzzy mathematics for product services services design

The concept of fuzzy sets was first proposed by Professor Zadeh at the University of California in 1965. Fuzzy sets have been applied in various fields, such as economics, science, and technology. They provide a new way to handle uncertainty and imprecision in an objective world, allowing for the analysis of uncertain judgments, extending the clarity and precision of mathematics into the fuzzy domain (Ma et al. 2021).

Shie et al. (2021) and Shie et al. (2022) applied fuzzy mathematics in a conceptual model for a new healthcare service design; whereas Shen and Wang (2016) and Wang, Yang, and Liu (2025) combined fuzzy mathematics with KE to explore new product forms that meet the demands of future customers by focusing on the most important issue of creativity in product form design. Kumru and Kumru (2013) and Hameed et al. (2022) applied fuzzy failure modes and impacted analysis to prioritise and evaluate possible failures during hospital emergency department work to improve patient confidence levels. Akgül et al. (2022) used KE to transfer customers' emotional needs into the formal elements of product design. Combined with fuzzy mathematics to correlate customers' emotional needs with the formal elements of cradle product design, they concluded that the language of fuzzy mathematics was effective for capturing customers' emotional needs (Li et al. 2021; Liu and Yang 2022; Tang et al. 2024).

The present study employed triangular fuzzy numbers to analyze the importance of perceptual vocabulary in children's digital reading book products, facilitating the analysis of consumer emotional needs, and reducing the level of customer dissatisfaction. Fuzzy hierarchical analysis effectively addresses the vagueness and uncertainty of perceptual vocabulary in product service functions, providing a critical approach to the sustainability of children's educational products.

3. Research framework: 4C-the proposed PSS design approach with abductive logic

This study proposed a comprehensive PSS design approach with abductive logic, titled '4C' model (combined from the initial letters of each phase), which is integrating KE, TRIZ, and Fuzzy techniques (as shown Figure 1). The research methodology consisted of four phases: Phase 1 was the analysis of consumer product attribute preferences, Phase 2 contained the

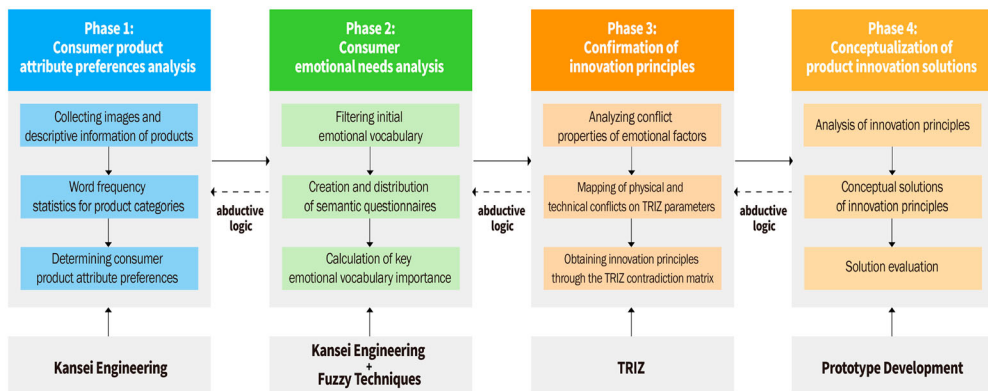


Figure 1. The proposed 4C design model.

analysis of consumer emotional needs, Phase 3 involved the confirmation of innovation principles, and Phase 4 contained the conceptualisation of product innovation solutions.

3.1. Phase 1: Consumer product attribute preferences analysis

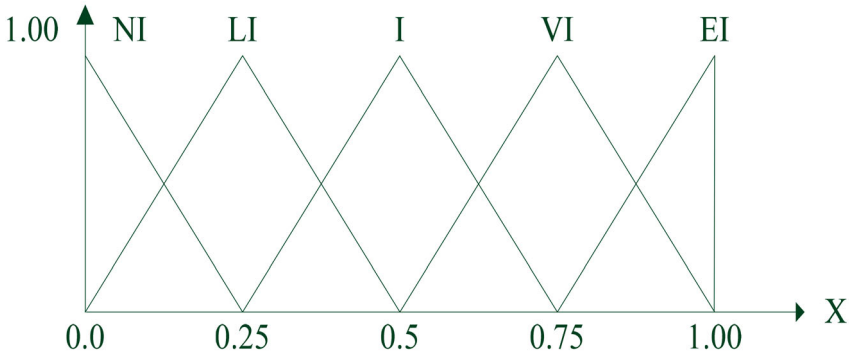
The purpose of this stage is to identify consumer preferences for product attributes, which will guide the subsequent collection of emotional vocabulary. First, images and descriptive information of mainstream products currently available on the market are collected to establish a product sample database. Next, conduct a frequency analysis and ranking of product category keywords found in the descriptive information and establish a database of similar product samples. Finally, form an expert panel to analyze and extract differences in product attributes from the database of similar products, thereby identifying consumer preferences for product attributes.

3.2. Phase 2: Consumer emotional needs analysis

The purpose of this stage is to obtain core emotional vocabulary and clarify consumers' core emotional needs. First, based on the consumer product attribute preferences identified in the first stage, collect corresponding emotional vocabulary from the product sample database, screen for key emotional vocabulary, and establish a key emotional vocabulary database. Next, conduct a five-point Likert scale survey based on the key emotional vocabulary to gather consumer opinions on products. Finally, use the fuzzy method for quantitative analysis to determine the core emotional vocabulary. To address the ambiguity and uncertainty of consumer opinions, apply fuzzy mathematics to the collected survey data for fuzzification. The specific steps are as follows:

- Fuzzification of questionnaire data

The fuzzy method offers a probability range and a five-point scale for the weighted importance, namely EI (Extremely Important), VI (Very Important), I (Important), LI (a Little Important), and NI (Not Important), and all membership functions for the semantic variables were standardised between the interval 0, 1. Thus, the semantic variables and five



$$EI(x) = \begin{cases} 4x - 3, & 0.75 < x < 1 \\ 1, & x = 1 \\ 0, & \text{others} \end{cases}$$

$$VI(x) = \begin{cases} 4x - 2, & 0.5 < x < 0.75 \\ 1, & x = 0.75 \\ -4x + 4, & 0.75 < x < 1 \\ 0, & \text{others} \end{cases}$$

$$I(x) = \begin{cases} 4x - 1, & 0.25 < x < 0.5 \\ 1, & x = 0.5 \\ -4x + 3, & 0.5 < x < 0.75 \\ 0, & \text{others} \end{cases}$$

$$LI(x) = \begin{cases} 4x, & 0 < x < 0.25 \\ 1, & x = 0.25 \\ -4x + 2, & 0.25 < x < 0.5 \\ 0, & \text{others} \end{cases}$$

$$NI(x) = \begin{cases} 1, & x = 0 \\ -4x + 1, & 0 < x < 0.25 \\ 0, & \text{others} \end{cases}$$

Figure 2. Triangular fuzzy number model and the membership functions.

corresponding fuzzy numbers corresponded to the five levels according to the scores from low to high: 1 corresponded to EI (0.75,1,1), 2 to VI (0.5,0.75,1), 3 to I (0.25, 0.5,0.75), 4 to LI (0,0.25,0.5), and 5 to NI (0,0,0.25). Triangular fuzzy numbers and membership functions are shown in Figure 2.

$$EI(x) = \begin{cases} 4x - 3, & 0.75 < x < 1 \\ 1, & x = 1 \\ 0, & \text{others} \end{cases}$$

$$VI(x) = \begin{cases} 4x - 2, 0.5 < x < 0.75 \\ 1, x = 0.75 \\ -4x + 4, 0.75 < x < 1 \\ 0, \text{others} \end{cases}$$

$$I(x) = \begin{cases} 4x - 1, 0.25 < x < 0.5 \\ 1, x = 0.5 \\ -4x + 3, 0.5 < x < 0.75 \\ 0, \text{others} \end{cases}$$

$$LI(x) = \begin{cases} 4x, 0 < x < 0.25 \\ 1, x = 0.25 \\ -4x + 2, 0.25 < x < 0.5 \\ 0, \text{others} \end{cases}$$

$$NI(x) = \begin{cases} 1, x = 0 \\ -4x + 1, 0 < x < 0.25 \\ 0, \text{others} \end{cases}$$

- Calculation of fuzzy number mean

Questionnaire data were analyzed using fuzzy mathematics for the priority calculation of fuzzy numbers. Assuming $S_{itj} = (q_{itj}, o_{itj}, p_{itj})$, and the triangular fuzzy number of the j_{th} participant assessed the importance of the correlation between the t_{th} customer need and i_{th} category of perceptual vocabulary determinants. S_{it} is defined as the fuzzy number mean of the i_{th} category for the perceptual vocabulary determinants of the t_{th} customer need, where n is the number of participants (Eq. (1)):

$$S_{it} = \frac{1}{n} \sum_{j=1}^n S_{itj}. \quad (1)$$

$S_{it} = (q_{it}, o_{it}, p_{it})$ was calculated according to Eqs (2–4):

$$q_{it} = \frac{1}{n} \sum_{j=1}^n q_{itj}, \quad (2)$$

$$o_{it} = \frac{1}{n} \sum_{j=1}^n o_{itj}, \quad (3)$$

$$p_{it} = \frac{1}{n} \sum_{j=1}^n p_{itj}. \quad (4)$$

- Defuzzification

It is recommended to present the output results as clear data because they are easier to interpret. The triangular fuzzy number re-centering method used by Shie et al. was applied

(Shie et al. 2021; Shie et al. 2022). Letting X represent the defuzzification value of the fuzzy number of each semantic column (q_{it} , o_{it} , p_{it}), the defuzzification value can be calculated according to Eq. (5):

$$X = \frac{q_{it} + o_{it} + o_{it} + p_{it}}{4} \quad (5)$$

Based on the results of the above steps, the key emotional vocabulary can be ranked by importance. The most important emotional vocabulary, as determined by this ranking, will be selected as the core emotional vocabulary. This serves as the foundation for discussing the determining factors of conflicts between product design and consumer emotional needs. Furthermore, the identified factors for improvement and deterioration will be matched with the corresponding TRIZ engineering parameters.

3.3. Phase 3: Confirmation of innovation principles

The purpose of this stage is to identify corresponding TRIZ innovation principles based on the core emotional vocabulary. Building on the core emotional vocabulary obtained in the second stage, a TRIZ contradiction matrix will be constructed to analyze the innovation principles that align with these emotional terms.

3.4. Phase 4: Conceptualisation of product innovation solutions

The purpose of this phase is to generate corresponding innovative solutions based on innovative principles, which are further expressed figuratively in the product design. Building on the TRIZ innovation principles established in the third phase, the expert team formulates functional solutions for children's digital reading book products. Meanwhile, consumers evaluate these solutions through a voting process, ensuring that all final service functional solutions align with user requirements.

4. Empirical case illustration

The Chinese government's 2021 Annual Work Report emphasised the promotion of the Reading for All strategy, especially among children. Amid technological advancements and the COVID-19 pandemic, digital reading for children has become a priority. However, most children's digital reading book products currently focus on consumer needs such as quality, price, and functionality, while overlooking the emotional needs of users (Furenes, Kucirkova, and Bus 2021). Children's cognitive and emotional development stages differ from those of adults, leading to unique preferences and needs for sensory elements such as colours, shapes, and sounds (Dresp-Langley 2020; König et al. 2022). Designing children's digital reading book products to address these emotional needs can enhance their reading interest and develop reading habits, ultimately improving societal educational and cultural literacy. Based on the policy and education environment, we choose children's digital reading book products as the case study to illustrate the proposed approach.

Table 1. Children's digital reading book products category keyword ratings.

Keywords	Frequency	Ranking
Pad	7	6
Early education machine	17	1
E-book device	11	4
Learning machine	13	3
AI robot	17	1
Add-in card	5	8
Story player	9	5
Point reading machine	14	2
Intelligence-enhancing machine	6	7

4.1. Phase 1: Consumer product attribute preferences analysis for Children's digital reading book products

Since consumers primarily purchase children's digital reading book products online, this study first collected 50 samples from major e-commerce platforms such as JD and Taobao, sorted by sales volume from high to low, to build a product sample database (Appendix A). Next, we conducted a word frequency analysis on the keywords related to product categories in the sample descriptions, with the results shown in Table 1. Based on principles of representativeness and comprehensiveness, this study selected the top six product categories from the word frequency analysis as representative products for children's digital readers, forming a representative product sample database as shown in Table 2. Finally, an expert group (including a product design director with 12 years of experience, a digital media designer with 10 years of experience, a digital network designer with 14 years of experience, and two graduate students specialising in industrial design and media design) discussed the distinctive features of these six different categories of children's digital reading book products based on the product descriptions and images from the representative product sample database. They identified five key product attributes: shape, function, material, emotional, and touch. This analysis highlighted that consumers prefer children's digital reading book products that are well-designed in these five attributes, guiding the subsequent product design based on perceptual needs.

4.2. Phase 2: Consumer emotional needs analysis for children's digital reading book products

4.2.1. Collection of emotional vocabulary and establishment of key emotional vocabulary database

In this study, we employed brainstorming techniques to gather emotional vocabulary, focusing on the five product attribute categories identified in the previous stage: shape, function, material, emotional, and touch. We aimed to collect as much perceptual demand information related to the design of children's digital reading book products as possible. This process resulted in 160 perceptual vocabulary terms. Due to space constraints, only a portion of these terms is displayed in Table 3.

Using group discussion methods, we screened the initial 160 emotional vocabulary terms, eliminating duplicates and synonyms to extract the most representative terms. This process yielded 20 key emotional vocabulary terms. In line with emotional research

Table 2. The representative product sample database.

No.	Category	Description	Product
1	Early education machine	✓ Cloud synchronisation support ✓ Customized learning programmes ✓ Voice interaction capability ✓ Touchscreen interface for intuitive and efficient operation	
2	AI robot	✓ AI large model integration ✓ Voice assistant with high-fidelity audio ✓ Accurate pronunciation for fluent English speaking ✓ Dual-mode connectivity with Wi-Fi and Bluetooth	
3	Point reading machine	✓ Touch-activated sound for enhanced interactivity ✓ Bilingual reading in Chinese and English ✓ Diverse illustrated patterns for increased engagement ✓ Rich content for educational and cognitive development ✓ Challenge-based mode for brain training	
4	Learning machine	✓ Compact and portable design ✓ Extensive knowledge database ✓ Online teaching and Q&A support ✓ Parental control functionality	
5	E-book device	✓ Ultra-clear e-ink display ✓ No backlight or adjustable front light ✓ Lightweight and pocket-sized ✓ Extended battery life ✓ Water-resistant design ✓ Large storage capacity with a vast collection of books	
6	Story player	✓ Six major early education story themes ✓ User-friendly operation ✓ Long-lasting playback ✓ Waterproof and drop-resistant ✓ Bluetooth speaker connectivity	

requirements, we employed the semantic differential method to pair each key term with its antonym, forming 20 pairs of key emotional vocabulary. Each pair, along with the corresponding intention images, was described and analyzed to establish a key emotional vocabulary database. Due to space constraints, only a portion of the key emotional vocabulary database is shown in Table 4.

Table 3. Emotional vocabularies for children's digital reading book products.

Children's digital reading book product perceptual vocabulary				
Shape	Cute	Fresh	Trendy	Cartoon
	Dumb	Smooth	Ugly	Light
Function	Entertaining	Convenient	Quick	Massive
	Simple	Natural	Safe	Hash
Material	Durable	Thicken	Fragile	Eco-friendly
	Robust	Drop-resistant	Healthy	Cleanable
Emotional	Comfortable	Companionable	Addictive	Enjoyable
	Active	Dull	Interactive	Indifferent
Touch	Comfortable	Grainy	Skin-friendly	Delicate

A five-point scale questionnaire was created on WenJuanXing using 20 pairs of key emotional vocabulary. The semantic differential method included paired adjectives with positive and negative meanings, and five intervals were set for each pair. The response intensity of words with opposing natures was reflected in the choice interval of words in the questionnaire, with positive semantics on the left, and negative semantics on the right. The lower the score, the closer the perceptual semantics of the left side.

Considering that parents and kindergarten teachers are the most knowledgeable about children, this study chose to distribute questionnaires through these two channels. We selected parents and teachers from Xingfuquan Kindergarten and Honghuanglan Kindergarten in Chaoyang and Haidian districts of Beijing, both of which emphasise children's reading and offer digital reading courses. A total of 250 questionnaires were distributed, and all 201 were successfully collected.

4.2.2. Construction of emotional vocabulary database based on fuzzy triangular mathematics

Based on the collected questionnaire results, this study calculates the importance of key emotional word pairs using triangular fuzzy mathematics to identify the core emotional vocabulary pairs. The key emotional word pairs in the questionnaire, rated on a scale of 1–5, were semantically converted. Using Eqs (1–4), the q_{it} , o_{it} , and p_{it} values for each pair were calculated. Defuzzification was performed using Eq. (5), which employed the center-of-gravity method of an integrated triangular fuzzy number. Based on the defuzzification results, the key emotional word pairs were ranked by importance, as shown in Table 5. The study selected the top four most important key emotional word pairs: 'safe – dangerous', 'delicate – rough', 'easy to clean – not easy to clean', and 'eye-protective – harsh light', as the core emotional vocabulary pairs.

4.3. Phase 3: Confirmation of innovation principles for children's digital reading book products

4.3.1. Analysis of conflict characteristics for emotional factors in children's digital reading book products

TRIZ conflicts are classified as physical and technical. Physical conflict states that to achieve a certain functional purpose, increasing a certain parameter characteristic of the technical system is necessary; however, it is necessary to reduce the parameter characteristics of technical systems for functional purposes. Technical conflict refers to an action that results

Table 4. Key emotional vocabulary database for children’s digital reading book products.

No.	Positive – negative perceptual vocabulary	Image	Description
1	Cute – Dumb		✓ Cartoon image ✓ Cute modelling ✓ Popular colours
2	Fashionable – Traditional		✓ Stylish ✓ Beautiful colour scheme ✓ Display of personality
3	Light – Heavy		✓ Lightweight ✓ Small size ✓ Slim and convenient
...	...	...	...
18	Recyclable – Non-recyclable		✓ Low carbon material ✓ Eco-friendly ✓ Low pollution ✓ Low emission
19	Easy to clean – Not easy to clean		✓ Easy to clean ✓ Not easy to get dirty
20	Delicate – Rough		✓ Gentle shell ✓ Comfortable keyboard ✓ Delicate material

in both useful and harmful outcomes, and the system concludes that several elements or parameters conflict with each other. In each subsystem, setting up a beneficial function causes another system to produce a harmful function; whereas eliminating a harmful function causes another system’s useful function to be damaged (Altshuller 1999). There is an interconnection between the separation and innovation principles for resolving physical and technical conflicts, respectively. There is correspondence between the four separation principles for physical conflicts and the 40 innovation principles, providing a quick way to

Table 5. Center-of-gravity method of integrated triangular fuzzy number and its important ranking.

Determinant	Integrated triangular fuzzy number			Defuzzied fuzzy number	Ranking importance
	q _{it}	O _{it}	P _{it}		
1	0.546	0.784	0.326	0.610	18
2	0.516	0.746	0.868	0.719	17
3	0.553	0.803	0.914	0.768	16
4	0.590	0.840	0.923	0.798	11
5	0.595	0.845	0.930	0.804	8
6	0.580	0.830	0.920	0.790	13
7	0.583	0.833	0.925	0.794	12
8	0.593	0.843	0.933	0.803	9
9	0.592	0.842	0.932	0.802	10
10	0.619	0.869	0.942	0.825	1*
11	0.595	0.845	0.935	0.805	7
12	0.578	0.828	0.922	0.789	14
13	0.595	0.845	0.935	0.805	7
14	0.583	0.833	0.925	0.794	12
15	0.576	0.826	0.925	0.788	15
16	0.607	0.857	0.938	0.815	4*
17	0.600	0.850	0.932	0.808	6
18	0.601	0.851	0.935	0.81	5
19	0.614	0.864	0.944	0.822	3*
20	0.616	0.867	0.944	0.824	2*

analyze conflicts for problem-solving. The two conflicts can be transformed into each other, and the corresponding relationships are listed in Table 6 (Geng et al. 2021).

Based on previous data analysis, this study applied four core emotional vocabulary pairs derived from product design elements. Physical conflicts were delicate – rough, easy to clean – not easy to clean; whereas technical conflicts were safe – dangerous, and eye-protective – harsh light.

One pair of physical conflicts was **delicate – rough**. If the product material is delicate, gentle, and the touch of the button is comfortable, problems will occur because the product is too smooth to carry, while more rough surfaces can hurt children easily. According to the four separation principles summarised by Altshuller for solving physical conflict, this study employed the space separation principle. For the two conflicting functions of the product material to be delicate and comfortable, but also rough and easy to carry, separated into different systems in space so that they can perform their functions without counteracting each other.

Another pair of physical conflicts was **easy to clean – not easy to clean**. If the product is easy to clean, its structure is simple, and the material is smooth. The realisation of this problem will lead to new problems, such as an inability to include more features on the surface of the shape. The choice of smooth material can also be difficult for children to grasp and fall easily. This study employed the principle of conditional separation by separating the frequently gripped and easily dirtied location as a condition separation, and changing the function of the system to resolve this conflict (Table 7).

One pair of technical conflicts was **safe – dangerous**. The process of pursuing product security causes the failure of certain functions. The goal is to ensure the security of the product network, the ability to control it remotely, and the safety of the material. The solution is to increase the functionality of product safety testing, and choose safe materials, which will

Table 6. Corresponding relationships between separation and innovation principles.

Separation principles			Innovation principles		
<i>Space separation</i>		1 Segmentation	2 Extraction, separation, removal, segregation	3	Local quality
		4 Asymmetry	7 Nested doll	13	Inversion, do it in reverse
		14 Spheroidality, curvilinearity	17 Transition into a new dimension	24	Mediator
		26 Copying	30 Flexible membranes or thin films		
<i>Time separation</i>		9 Prior counteraction	10 Prior action	11	Cushion in advance, prior counteraction
		15 Dynamicity, optimisation	16 Partial or excessive action	18	Mechanical vibration
		19 Periodic action	20 Continuity of useful action	21	Rushing through
		29 Pneumatic or hydraulic construction	34 Rejecting and regenerating parts	37	Thermal expansion
<i>Separation of whole and part</i>	Conversion to subsystems	1 Segmentation	12 Equipotentiality, remove stress	25	Self-service, self-organisation
		33 Homogeneity	40 Composite materials		
	Conversion to supersystems	5 Combining, integration, merging	6 Universality, multi-functionality	22	Convert harm into benefit
	Conversion to competitive systems	23 Feedback			
<i>Conditional separation</i>	Conversion to inverse systems	27 Disposable			
		13 Inversion, do it in reverse	8 Counterweight		
		28 Replacement of mechanical system	29 Pneumatic or hydraulic construction	31	Porous material
		32 Changing the colour	35 Transformation of properties, parameter change	36	Phase transition
		38 Accelerated oxidation, enriched atmospheres	39 Inert environment		

raise new problems and increase product costs. By checking the 39 engineering parameters of TRIZ, those that required improvement were serial number 27 for reliability and serial number 37 for difficulty in detection/measurement.

The secondary pair of technical conflicts was **eye-protective – harsh light**. The two key stimulating factors of the screen on eyes are distance and light. When the distance is too short, visual fatigue occurs when the eyes are not resting. High screen brightness is harsher than natural light, which is harmful to the eyes. Solving eye health problems requires lowering the brightness of the screen and increasing the adjustment distance; however, this new problem reduces the saturation and clarity of the product screen. Checking the 39 technical engineering parameters of TRIZ, the parameter serial number 18 needs to be improved,

Table 7. Conversion of physical conflicts into service function solutions based on TRIZ.

Conflict property	Perceptual conflict vocabulary	Separation principle	Description
Physical conflicts	Delicate – Rough	Space separation	The two conflicting functions of the product material (delicate or rough but easy to carry) are separated into different systems in space. Let these two systems perform their tasks without contradicting each other.
	Easy to clean – Not easy to clean	Conditional separation	The product needs to be easy to clean and have a simple shape, separating the frequently gripped, easily dirtied position as a condition, changing the function of this part of the system,

Table 8. Conversion of physical conflicts into service function solutions based on TRIZ.

Conflict property	Perceptual conflict vocabulary	Improved parameters	Relative deteriorated parameters
Technical conflicts	Safe – Dangerous	27 Reliability: the ability of the system to complete the specified function in the specified method and state.	37 Difficulty in detection/measurement: Difficulty in monitoring the system, which takes a long time to build and use due to the complexity and high cost of the system, complex relationships between components, high testing accuracy, and increased costs of testing.
	Eye-protective – Harsh light	18 Luminous intensity refers to the amount of luminous flux per unit area, and the light characteristics of the system, such as brightness and quality.	24 Loss of information, defined as partial or total, permanent, or temporary loss of data.

Table 9. Results of the corresponding relationship between the separation principle of physical conflict and the innovation principle.

Perceptual vocabulary	Separation principles	Innovation principles
<i>Delicate – Rough</i>	Space separation	2, 3
<i>Easy to clean – Not easy to clean</i>	Conditional separation	32, 35

addressing the luminous intensity. Then, the corresponding worsened parameter was serial number 24, addressing information loss. Table 8 shows the solution to the technical conflicts transformed into a service function for safe – dangerous and eye-protective – harsh light, as defined by Altshuller (1997).

4.3.2. Mapping of physical and technical conflicts on TRIZ parameters

According to the Table 7 of the corresponding relationships between the separation principle of physical conflict and the innovation principle, that corresponding to delicate – rough was (2, 3), while the separation principle corresponding to the easy-to-clean – not easy to clean is the conditional separation, relating to the innovation principle (32, 35) (Table 9).

Table 10. Corresponding results of the TRIZ engineering parameter matrix.

No.	No.	37	24
No.	Deteriorated parameters		
Improved parameters	Difficulty in detection/measurement	Loss of information	
27	Reliability	27, 40, 28	
18	Luminous intensity		1, 6

According to the 39 contradictory parameters of TRIZ, the vertical columns are the parameters that require improvement from serial numbers 1–39, while the horizontal columns are those that have deteriorated from serial numbers 1–39. A specific matrix element is obtained at the intersection of two serial numbers, and the number given by that element is the recommended serial number of the innovation principle. It was found that the parameter to be improved for safe – danger was serial number 27, and deteriorated was serial number 37, corresponding to engineering parameters (27, 40, 28). Comparatively, the parameter to be improved for eye-protective – harsh light was serial number 18, and deteriorated was serial number 24, corresponding to engineering parameters (1, 6) (Table 10).

4.3.3. Innovation principle proposed based on TRIZ contradiction matrix analysis

According to the previous phase, the following innovation principles were found: the corresponding innovation principles for delicate were 2 and 3, the corresponding innovation principles for easy to clean were 32 and 35, while the corresponding engineering parameter matrices for safe were 27, 40, and 28, and those for eye protection were 1 and 6. Table 11 presents the definitions of these TRIZ innovation principles and their corresponding product innovation design concepts.

4.4. Phase 4: Conceptualisation of product innovation solutions for children's digital reading book products

4.4.1. Service functional solutions for children's digital reading book products

We re-engaged the expert panel from the first phase, tasking them with proposing service functional solutions for children's digital reading book products based on the innovation principles corresponding to each core emotional need. Additionally, we invited 10 parents and teachers from Xingfuquan Kindergarten and Honghuanglan Kindergarten in Chaoyang and Haidian districts of Beijing to provide feedback on these solutions.

In this process, the expert panel first proposed the service functional solutions, which were then evaluated by parents and teachers through a voting mechanism. Solutions receiving eight or more votes were considered to meet the emotional needs, whereas those receiving fewer than eight votes required further refinement based on the feedback until they met the voting threshold. Ultimately, we identified nine service functional solutions for children's digital reading book products.

1. SF₁: The product body is made of wear – and UV-resistant material, and only the part children grasp can be made of silicone that can reach the edible level.

Table 11. Definitions of TRIZ innovation principles and their corresponding product innovation design concepts.

Innovation principles			Definitions of TRIZ innovation principles and their corresponding product innovation design concepts
Physical conflicts	<i>Space separation</i>	2 Extraction, separation, removal, segregation	Refers to the part or property that produces a negative impact extracted from the product. The device is smooth and inconvenient for children to grip, so it is possible to separate the part of the product that has a negative impact (i.e. different parts of the product meet different service function requirements).
		3 Local quality	Means that different parts of the product fulfil different functions. The material of the device body should be smooth and delicate, but also convenient for children to grasp. It can change the product's partial material to facilitate children.
	<i>Conditional separation</i>	32 Changing colour	This principle is to change the colour of the object itself. The main users are children, who are more sensitive to bright colours, but easily dirty the device. Therefore, changing the colour to a less dirty colour in the place where children often grasp is recommended.
		35 Transformation of properties, parameter change	It is possible to change the shape or flexibility of the product. The device material of the button and grip position can be changed partially to facilitate cleaning.
Technical conflicts	27 Disposable		Factors that affect the safety can be dropped. Harmful online content can negatively impact children's psychological development; therefore, it is essential to filter out such inappropriate content.
	40 Composite materials		Children's supplies have high requirements for materials to ensure that children are not harmed. Younger children like to chew; thus, products must use composite materials.
	28 Replacement of mechanical system		Changing the fixed field into a mobile field, and the static into a dynamic. This can be improved by examining and enhancing parental supervision ways for children's use of the product.
	1 Segmentation		This principle refers to dividing a system, object, or process into smaller, independent parts. Most devices use ink screens without blue light to protect children's eyes. According to this innovative principle of separation, the eye protection function can be separated into different items.
	6 Universality, multi-functionality		Refers to designing a component or system to perform multiple functions. It can be equipped with a usage time limit and a fatigue reminder function to support eye protection.

- SF₂: The parts that children often grasp are made of wear-resistant patterns to facilitate their grip so that different parts complete different functions.
- SF₃: Change the colour of the parts often grasped to brown; make them easy to clean.

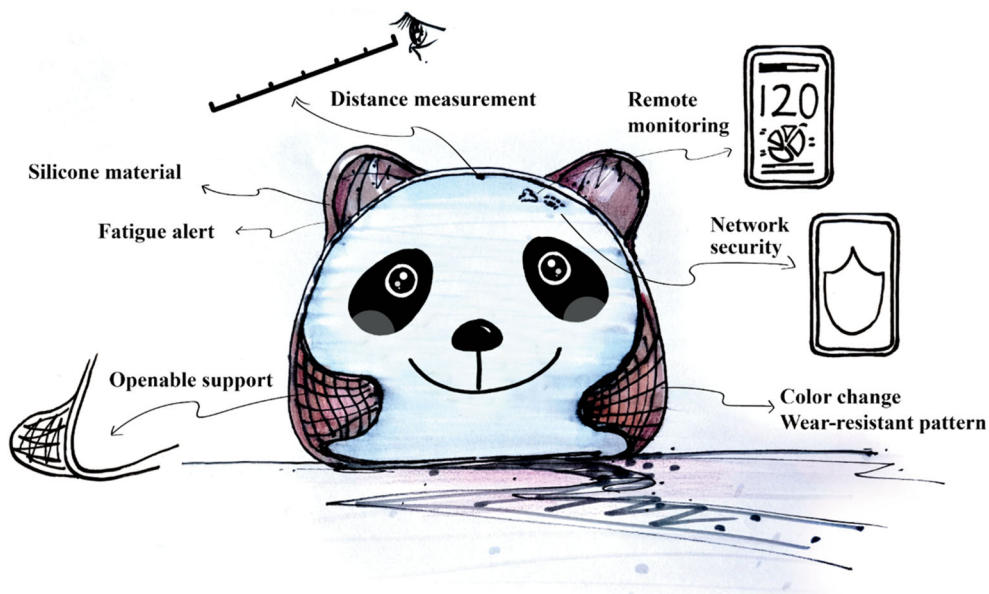


Figure 3. Sketch of children's digital reading book product.

4. SF₄: Change the material of the part that children often grasp, using silicone material, with a removable function, easy to clean, and folded for support.
5. SF₅: AI technology blocks harmful online content and provides personalised recommendations.
6. SF₆: Younger age groups of children like to chew. To be safe, choose a safe material, which can reach the edible level of silicone.
7. SF₇: Provide the function of connecting with cell phones through a small programme or app to achieve the purpose of parental remote control.
8. SF₈: For eye protection, in addition to retaining the ink screen, a camera can also be added to measure distance. When the child is too close, the screen will automatically remind the child to increase their distance.
9. SF₉: The product is installed with a screen use limit and fatigue reminder. When screen use has reached a certain duration, the ear area in the design draft turns red, and the child is reminded to take a break.

4.4.2. Conceptualisation of service functional solutions

Figure 3 shows a draft of the children's digital reading book products, Figure 4 shows a 3D view of the solution, and Figure 5 shows a simulation using 3D modelling. The modelling source is the panda image, which is popular among children, and the service function solution was converted into a product design.

Figure 6 shows the interface of the remote-control app for parents. Parents can use the app to know the children's daily learning time, how many books they have read, the type and corresponding proportion of books read, and a summary of the children's learning by stage. Meanwhile, this app uses AI technology to block harmful content and recommend personalised content suitable for children.

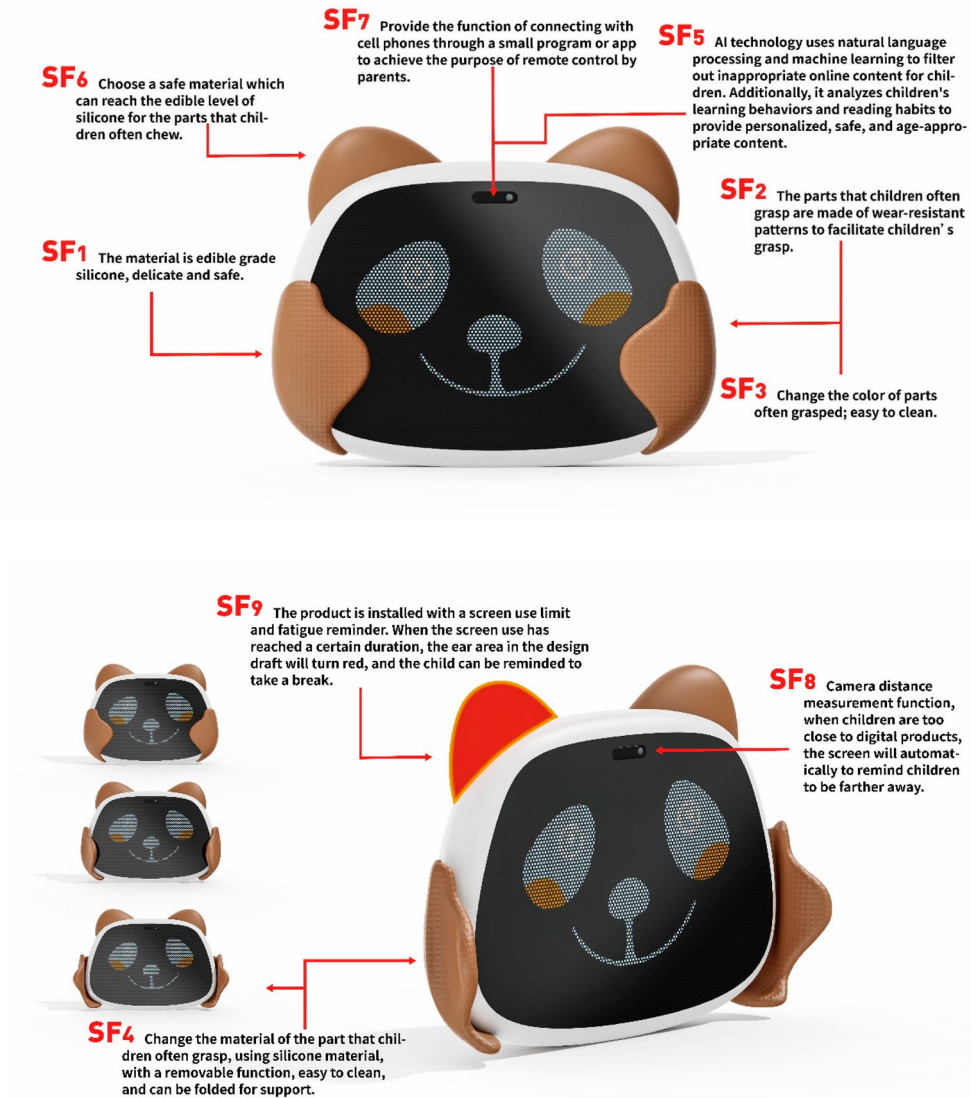


Figure 4. 3D view of proposed solutions.

4.4.3. Prototype development and testing

The prototype development in this study was technically supported by Beijing Yiqi Design Co., Ltd., utilising Digital Twin Technology for product simulation and interaction testing to optimise the conceptual design of the children's digital reading book. During the prototype development process, particular emphasis was placed on material selection, interactive functionality, intelligent control mechanisms, and eye protection features, ensuring the scientific rigour, feasibility, and enhanced user experience of the product design.

Initially, a high-precision digital model was constructed using 3D CAD modelling, and different material compositions were evaluated in a virtual physical environment to assess grip



Figure 5. 3D concept model.

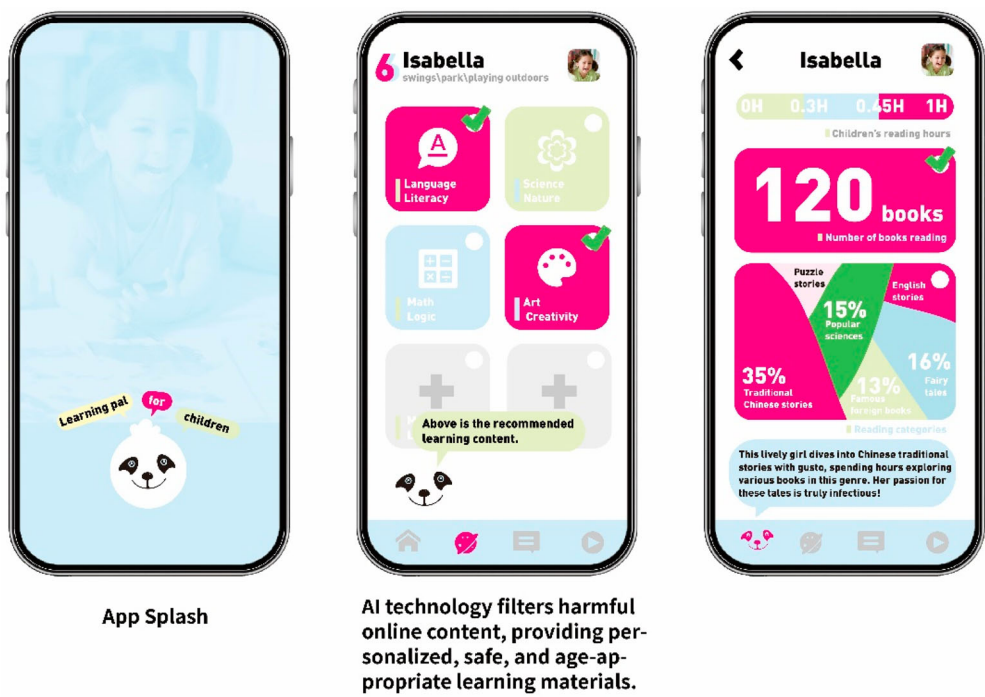


Figure 6. User interface of parental remote control.

comfort and durability. Experimental results demonstrated that, compared to conventional plastic materials, using food-grade silicone (SF₁, SF₄, SF₆) for the grip area significantly improved anti-slip performance and exhibited superior stain resistance, reducing stain retention by approximately 35%. Further simulation analysis revealed that the anti-slip texture design (SF₂) provided adequate grip stability in over 90% of usage scenarios, thereby enhancing the ergonomic properties of the product and improving grip comfort and safety during use.

Table 12. The satisfaction analysis for the proposed service functions of a digital reading product for children.

Service Functions (SF)	Satisfaction (in percentage)				
	Very Dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied
SF ₁ , SF ₆	0%	0.87%	12.17%	24.35%	62.61%
SF ₂	0%	0%	9.57%	23.48%	66.96%
SF ₃ , SF ₄	0%	0.87%	10.43%	25.22%	63.48%
SF ₅	0%	0%	9.57%	28.70%	61.74%
SF ₇	0%	0%	8.70%	23.48%	67.83%
SF ₈	0%	0%	9.57%	26.09%	64.35%
SF ₉	0%	0%	8.70%	27.83%	63.48%
Overall Satisfaction	0%	0%	10.43%	29.57%	60.00%

For intelligent function evaluation, a personalised AI-based recommendation system (SF₅) was developed and trained using 100 sets of virtual user behaviour data to assess content matching accuracy. Experimental results showed that the system achieved an average recommendation accuracy of 87.6%, effectively identifying children's reading preferences and providing personalised book recommendations. Additionally, the system integrated an intelligent content filtering mechanism, which autonomously blocks inappropriate online content, ensuring a safe reading environment for children. In evaluating the remote parental management function (SF₇), simulated parental interactions demonstrated that the system's average response latency was below 250 milliseconds, meeting real-time management requirements. Testing of the remote control interface showed that parents could efficiently adjust reading duration, content access permissions, and filtering levels, enhancing both reading security and individualised content management for children.

Regarding the eye protection features (SF₈, SF₉), 200 instances of children's close-range reading behaviours were simulated to test the effectiveness of the camera-based distance monitoring system. The results indicated that the system successfully detected over 95.5% of cases where children were reading at an excessively close distance, triggering appropriate distance reminders. Additionally, the screen usage limit and fatigue reminder function accurately monitored prolonged reading durations, issuing timely rest notifications within the predefined time thresholds to effectively reduce eye strain caused by extended close-range screen exposure.

To further illustrate the functionality and feasibility of the proposed design, we have produced a final prototype as the proof of concept (POC) demonstration video, which is available at the following link: (https://www.bilibili.com/video/BV1jZ9XYpE81/?spm_id_from=333.337.search-card.all.click&vd_source=fd090fdee4dcae27d2b15734e49c0219).

4.4.4. Evaluation

This study conducted a conceptual design of a prototype digital reading product for children. Continuous monitoring of consumer demand is required to test whether it satisfies the needs of consumers. A questionnaire was distributed for data collection. Due to the redundancy and similarity in design objectives among certain solutions, SF₁ and SF₆, as well as SF₃ and SF₄, were combined for analysis. The target group consisted of product managers, designers, and users who used the children's reading products. A total of 115 questionnaires were collected, and the results are presented in Table 12.

Table 13. Results of multiple linear regression.

Variables	Regression Coefficient (β)	Standard Error (SE)	t-value	p-value
SF ₁ &SF ₆	0.21	0.08	2.63	0.011
SF ₂	0.18	0.07	2.57	0.014
SF ₃ &SF ₄	0.12	0.09	1.33	0.187
SF ₅	0.25	0.07	3.57	0.002
SF ₇	0.27	0.06	4.11	< 0.001
SF ₈	0.22	0.07	3.14	0.005
SF ₉	0.15	0.08	1.88	0.072
Constant	1.87	0.34	5.50	< 0.001

Note: In multiple linear regression analysis, β represents the magnitude of each independent variable's impact on the dependent variable (Mason and Perreault 1991).

Based on the user satisfaction survey results, all service functions received high recognition in user experience, with the combined percentage of 'satisfied' and 'very satisfied' exceeding 85%, indicating that the overall product design aligns well with user needs.

Furthermore, multiple linear regression analysis was employed to investigate the impact of these service functions on overall user satisfaction. The overall satisfaction rating was used as the dependent variable, while the satisfaction ratings of each service function (the percentage of 'satisfied' and 'very satisfied' responses) were treated as independent variables. The regression model was constructed as follows:

$$\text{Overall Satisfaction} = \beta_0 + \beta_1 \text{SF}_1 \& \text{SF}_6 + \beta_2 \text{SF}_2 + \beta_3 \text{SF}_3 \& \text{SF}_4 + \beta_4 \text{SF}_5 + \beta_5 \text{SF}_7 \\ + \beta_6 \text{SF}_8 + \beta_7 \text{SF}_9 + \varepsilon$$

Here, β_i represents the regression coefficients, indicating the contribution of each service function to overall user satisfaction, while ε denotes the error term. As shown in Table 13, the regression analysis results indicate that the model has a coefficient of determination $R^2 = 0.87$, suggesting that the satisfaction ratings of the service functions explain 87% of the variance in overall satisfaction. This demonstrates a strong model fit ($p < 0.001$).

According to the results in Table 13, SF₁&SF₆, SF₂, SF₅, SF₇, and SF₈ were found to have a significant impact on user satisfaction (p -value < 0.05). Among them, SF₇ contributed the most to the improvement of overall user satisfaction, followed by SF₅ and SF₈ ($\beta_{(\text{SF}_7)} > \beta_{(\text{SF}_5)} > \beta_{(\text{SF}_8)} > \beta_{(\text{SF}_1 \& \text{SF}_6)} > \beta_{(\text{SF}_2)}$), indicating that these three design features best align with user needs and significantly enhance the user experience. Compared to existing market products, these service functions exhibit notable advantages in both functionality and intelligence.

SF₇ enables parents to remotely manage the children's digital reading book product by connecting to a mobile application, allowing real-time adjustments to reading duration, content access permissions, and monitoring of children's reading habits. Most children's reading devices on the market only support basic features such as timed screen locks or static content whitelists, lacking flexible and real-time control capabilities. In contrast, the remote management function proposed in this study offers greater interactivity and convenience, allowing parents to dynamically adjust usage rules based on their child's needs in different scenarios. Some surveyed parents mentioned that this feature allows them to more precisely monitor their child's reading behaviour and intervene promptly when necessary, without relying on the child to self-regulate their screen time. This function has been

particularly well-received by working parents, who appreciate how remote control makes it easier to integrate the digital reading device into the family's educational management system.

SF₅ leverages AI technology to optimise personalised recommendations and content safety management. Most children's digital reading book products on the market still rely on keyword matching or basic classification algorithms, resulting in lower recommendation accuracy and requiring parents to manually filter suitable reading materials for their children. In contrast, the AI recommendation system proposed in this study dynamically adjusts recommended content based on children's reading habits and interests, enhancing their willingness to independently explore new books. Additionally, the AI filtering system automatically blocks inappropriate content, reducing the need for parental intervention. User feedback indicates that many parents observed their children becoming more inclined to explore new books rather than repeatedly reading a fixed selection of content. Furthermore, parents expressed high approval of the system's content safety mechanisms, acknowledging its ability to provide a controlled yet personalised reading environment for children.

SF₈ integrates e-ink display technology with a camera-based distance measurement system to achieve intelligent vision protection. Currently, most children's digital reading book products rely on fixed-time reminders, which prompt users to take breaks at predetermined intervals. In contrast, the distance measurement system in this study's design continuously monitors the child's distance from the screen and provides real-time alerts when the child is too close. This feature aims to mitigate poor eye habits caused by prolonged, highly focused reading. User feedback indicates that children became more aware of their posture while reading and gradually developed healthier viewing distances. Additionally, some parents reported that this function reduced their need to frequently remind their child to 'keep a proper distance,' thereby improving the overall parent-child reading interaction experience.

In comparison, SF₃&SF₄ and SF₉ did not reach statistical significance in their impact on overall user satisfaction. This suggests that user evaluations of these features were relatively balanced – while they did not significantly enhance overall satisfaction, they also did not negatively affect the user experience. From the perspective of user perception, the improvements in detachable structure and colour optimisation (SF₃&SF₄) may be more related to personalisation preferences rather than directly influencing core usability, leading to varying levels of importance among different user groups. Regarding screen time reminder functionality (SF₉), although it plays a positive role in eye protection, users may prioritise real-time distance alerts (SF₈) over fixed-time usage restrictions, making the perceived importance of this feature more dependent on individual usage habits.

5. Conclusions

This method integrates Kansei Engineering (KE), the Theory of Inventive Problem Solving (TRIZ), and Fuzzy techniques to effectively manage design knowledge in the innovation of product conceptualisation and design. It offers a traceable pathway that links customer concerns, business context, and technological digitisation through abductive logic. An innovative case study of PSS-oriented children's educational product design is chosen

and presented to verify the proposed method. In this case, six product categories were identified, and their functions were described through interviews and related materials.

This study makes several significant academic contributions to the field of Product Service Systems (PSS) design, particularly from the novelty of integrating Kansei Engineering (KE), the Theory of Inventive Problem Solving (TRIZ), and Fuzzy techniques, as well as the application of abductive logic. These contributions are echoed through the key findings of the research:

1. Integrated Approach:

- **Comprehensive Methodology:** The study presents a novel and comprehensive approach to product conceptualisation and design by integrating KE, TRIZ, and Fuzzy techniques. This integration provides a robust framework that addresses both the functional and emotional aspects of product design, thereby enhancing the overall user experience.
- **Abductive Logic:** The application of abductive logic in the design process offers a structured yet flexible approach to problem-solving. This method allows designers to explore multiple possibilities and generate innovative solutions that are not only technically feasible but also emotionally resonant with users.

2. Emotional Needs:

- **User-Centric Design:** By emphasising the emotional needs of users through KE, the study highlights the importance of user-centric design in PSS. This approach ensures that products are not only functional but also emotionally engaging, leading to higher user satisfaction.
- **Enhanced User Satisfaction:** The findings demonstrate that addressing emotional needs significantly enhances user satisfaction. This contribution underscores the value of integrating emotional intelligence into the design process, thereby bridging the gap between user expectations and product capabilities.

3. Innovative Solutions:

- **TRIZ-Informed Innovation:** The study leverages TRIZ principles to generate innovative solutions for complex design problems. This application of TRIZ in the context of PSS design provides a systematic approach to resolving technical contradictions and fostering creativity in product development.
- **Effective Problem-Solving:** The integration of TRIZ with KE and Fuzzy techniques results in effective problem-solving strategies that are both practical and innovative. This contribution highlights the potential of TRIZ to drive innovation in PSS design, particularly in addressing the emotional and functional needs of users.

4. Consumer Satisfaction:

- **High Consumer Satisfaction:** The proposed method results in high consumer satisfaction, validating its effectiveness in addressing user needs and preferences. This finding underscores the importance of a holistic design approach that combines multiple methodologies to achieve optimal PSS design.
- **Empirical Validation:** The empirical case study of children's digital reading book products provides concrete evidence of the effectiveness of the proposed method. This validation through real-world application strengthens the academic contribution by demonstrating the practical applicability of the integrated approach.

To address RQ1, the study presents a comprehensive PSS design approach integrating KE, TRIZ, and Fuzzy techniques. This methodology provides a robust framework for addressing both functional and emotional needs in product design. The integration of these methodologies offers a structured yet flexible approach to product conceptualisation and design. KE addresses user emotional needs, TRIZ facilitates innovative solutions for complex design problems, and Fuzzy techniques manage design uncertainties. This integration is demonstrated through an empirical case study on children's digital reading book products, enhancing the overall design process.

For RQ2, the study emphasises user-centric design by incorporating emotional needs through KE. Focusing on emotional and sensory aspects, KE ensures products meet both functional requirements and user emotional needs, particularly for children. This approach significantly enhances user satisfaction, validated through an empirical case study showing higher user engagement and satisfaction with digital reading products.

Regarding RQ3, the study employs abductive logic to link customer concerns, business context, and technological digitisation in the design process. Abductive logic enables designers to explore multiple possibilities and generate solutions that are both technically feasible and emotionally resonant. Integrating abductive logic with KE, TRIZ, and Fuzzy techniques provides a structured problem-solving approach for modern product design. This ensures a user-focused design process, meeting both functional and emotional needs. The empirical validation through the case study of children's digital reading book products demonstrates the practical effectiveness of this integrated approach.

The proposed methodology holds significant value for driving innovation in digital reading products. To ensure its effective implementation and expand its impact, this study outlines two key directions for promotion. First, to enhance its applicability, a digital design assistance system can be developed by integrating KE, TRIZ, and Fuzzy techniques, aiming to improve the efficiency of the abductive reasoning process and achieve automation in user emotional demand analysis and innovative design. This system can be embedded into CAD and Product Lifecycle Management (PLM) software, enabling enterprises to efficiently generate optimised solutions during the product design process and accelerating the innovation of smart reading products. Second, by establishing industry standards and implementing policy guidance, user emotion-driven product innovation can be incorporated into the design norms for educational technology and smart reading devices. Additionally, leveraging academic conferences and industry alliances to promote this methodology will enhance its recognition in the field, providing theoretical and practical support for the continued innovation of digital reading products.

This study, while comprehensive, has limitations that suggest areas for future research. First, its focus on children's digital reading book products restricts the generalizability of findings to other industries. Future studies could test the integrated KE, TRIZ, and Fuzzy techniques in diverse contexts like healthcare or transportation to confirm broader applicability. Second, empirical validation through a single case study may not capture market segment complexities. Future research could use multiple case studies or large-scale investigations to provide robust evidence of the methodology's effectiveness. Third, reliance on abductive logic, while flexible, may not always yield definitive solutions. Future studies could integrate other logical frameworks to enhance design process robustness. Lastly, the study's emphasis on emotional needs through KE could be expanded to include usability and accessibility. Future research could incorporate these aspects to create more

inclusive designs. In summary, future research could address these limitations by exploring broader applications, conducting extensive empirical validations, integrating additional frameworks, and incorporating other user-centric principles.

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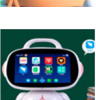
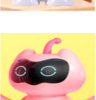
Appendix A

Table A1. The product sample database.

No.	Products	Descriptions	No.	Products	Descriptions
1		Intelligent learning tablet, simultaneous reading, professional tutors, cloud classroom	11		Early education pad robot, simultaneous English point reading machine
2		Children's learning pad, early education, scientific enlightenment	12		AI robot, TV learning machine, intelligence-enhancing enlightenment
3		Children's pad, eye protection screen, parent APP	13		Early Learning Storyteller, Point-and-Read Learning Machine, Robot
4		Student pad, the question bank	14		Bedtime story early education machine, projector, Bluetooth
5		Smart learning pad, large screen, intelligent homework tutoring	15		Intelligent early education machines, traditional Chinese children's songs, intelligence-enhancing learning
6		Smart learning pad, textbook tutoring	16		Early education machine, baby intelligent storytelling robot, children's music player
7		Early learning machine, point reading, story player, karaoke	17		Children's storytelling machine, audiobooks
8		Early education machine, intelligence-enhancing story, Intelligent robot learning machine	18		Children's early education machine, storytelling machine, intelligence-enhancing robot

(continued)

Table A1. Continued

No.	Products	Descriptions	No.	Products	Descriptions
9		Children's learning machine, early learning machine, English point reading machine	19		Portable speakers, pluggable cards.
10		Point reading, learning machine, AI robot	20		Card-based early education machine, bilingual
21		Children's early education, logical thinking training machine	31		Intelligent accompanying robot, early learning machine, storytelling machine
22		Children's early education machine, point reading, insert card learning	32		Educational story robot learning machine
23		Insert card early education machine, bilingual literacy, phonics learning machine	33		Colour ink screen reader
24		Early education card machine, card inserted bilingual enlightenment, point reading machine	34		E-book device
25		Early education card machine, insert card English learning machine, point reading	35		E-book device
26		AI robot, learning machine, point reading machine	36		Colour ink screen reader
27		AI robot, learning machine, point reading machine	37		Colour ink screen reader
28		AI robots, early education machines, children's music player	38		Ink screen reader
29		Children's story player, music player	39		E-book device
30		Early education machine, children's music player, storytelling machine	40		E-book device

(continued)

Table A1. Continued

No.	Products	Descriptions	No.	Products	Descriptions
41		Thin and light e-book reader	46		Picture book reading robot, point reading pen, story machine
42		Smart Handwriting e-book Reader	47		English learning machine, vocabulary e-learning
43		Smart e-book reader	48		Intelligent robot, early learning machine, point reading machine
44		Robot learning machine	49		Eye-care learning robot, finger-reading
45		AI learning machine, point reading, early education	50		Children's picture book reading robot, early learning machine